Simulation-based verification of bfpwr

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Package version 0.3
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This report compares **bfpwr** predictions with simulations across the listed priors, evidence thresholds, and sampling schedules.

1 z-Test

Simulated Design Priors

Table 1: Simulated design priors for the z-Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions	USD
Z1	Point(0)	10 to 500 by 10	10000	1
Z2	Point(0.2)	10 to 500 by 10	10000	1
Z3	Point(0.5)	10 to 500 by 10	10000	1
Z4	Normal(0.5, 0.1)	10 to 500 by 10	10000	1
Z5	Normal(0.2, 0.3)	10 to 500 by 10	10000	1
Z6	Normal(0.2, 0.5)	10 to 500 by 10	10000	1
Z7	Normal(0, 0.5)	10 to 500 by 10	10000	1
Z8	Point(-0.3)	10 to 500 by 10	10000	1
Z9	Point(0)	10 to 500 by 10	10000	50
Z10	Point(10)	10 to 500 by 10	10000	50
Z11	Normal(10, 20)	10 to 500 by 10	10000	50
Z12	Point(0)	10 to 10000 by 10	10000	1
Z13	Point(0.2)	10 to 10000 by 10	10000	1
Z14	Normal(0.2, 0.1)	10 to 10000 by 10	10000	1
Z15	Normal(0.2, 0.5)	10 to 10000 by 10	10000	1

Analysis Priors

Table 2: Analysis priors used for the z-Test verification run. Designs lists the matching short IDs from the design-prior table.

Function	H1	H0	Designs
bf01	Normal_+(-0.2, 0.3)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal_+(0, 0.1)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal_+(0, 0.707)	Point(0)	Z1-Z5, Z7-Z8, Z12-Z13
bf01	Normal_+(0, 2)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal_+(0.2, 0.3)	Point(0)	Z1-Z5, Z7-Z8

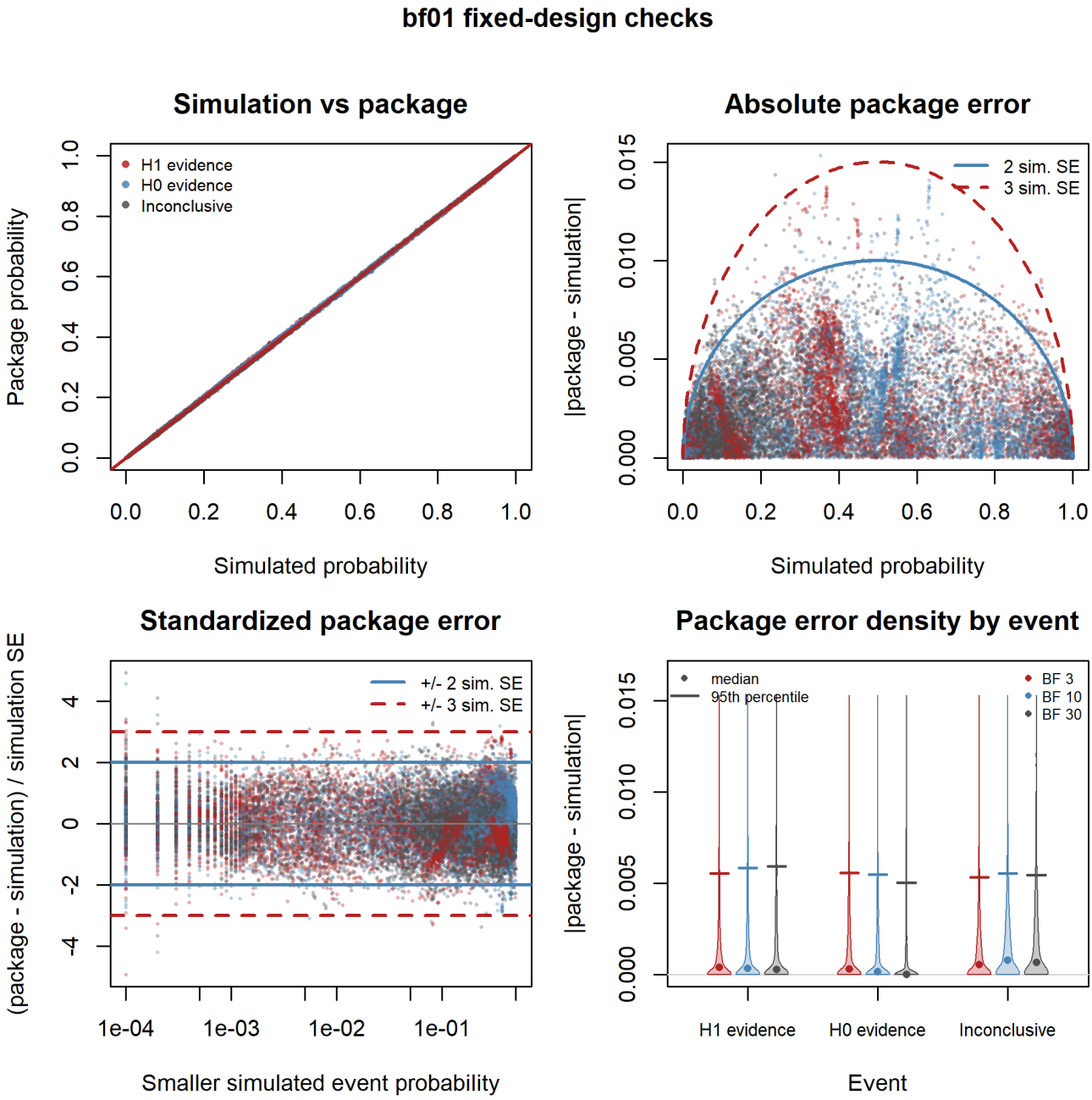
bf01	Normal$_-$(-0.2, 0.3)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal$_-$(0, 0.1)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal$_-$(0, 0.707)	Point(0)	Z1-Z5, Z7-Z8, Z12-Z13
bf01	Normal$_-$(0, 2)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal$_-$(0.2, 0.3)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal(0, 0.707)	Point(0)	Z1-Z8, Z12-Z15
bf01	Normal(0, 20)	Point(0)	Z9-Z11
bf01	Normal(0.2, 0.3)	Point(0)	Z1-Z8, Z12-Z15
bf01	Normal(0.2, 0.707)	Point(0.2)	Z2-Z6, Z13-Z15
bf01	Normal(10, 20)	Point(0)	Z9-Z11
bf01	Point(0.2)	Point(0)	Z1-Z8, Z12-Z15
bf01	Point(0.5)	Point(0)	Z1-Z8, Z12-Z15
bf01	Point(0.5)	Point(0.2)	Z2-Z6, Z13-Z15
bf01	Point(10)	Point(0)	Z9-Z11
dirbf01	Normal$_+$(0, 0.707)	Normal$_-$(0, 0.707)	Z1-Z8, Z12-Z15
dirbf01	Normal$_+$(0, 20)	Normal$_-$(0, 20)	Z9-Z11
dirbf01	Normal$_+$(0.2, 0.707; > 0.2)	Normal$_-$(0.2, 0.707; ≤ 0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment$_+$(0.354)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment$_+$(0.354; null = 0.2)	Point(0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment$_+$(0.707)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment$_+$(7.071)	Point(0)	Z9-Z11
nmbf01	Moment$_-$(0.354)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment$_-$(0.354; null = 0.2)	Point(0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment$_-$(0.707)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment$_-$(7.071)	Point(0)	Z9-Z11
nmbf01	Moment(0.354)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment(0.354; null = 0.2)	Point(0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment(0.707)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment(7.071)	Point(0)	Z9-Z11

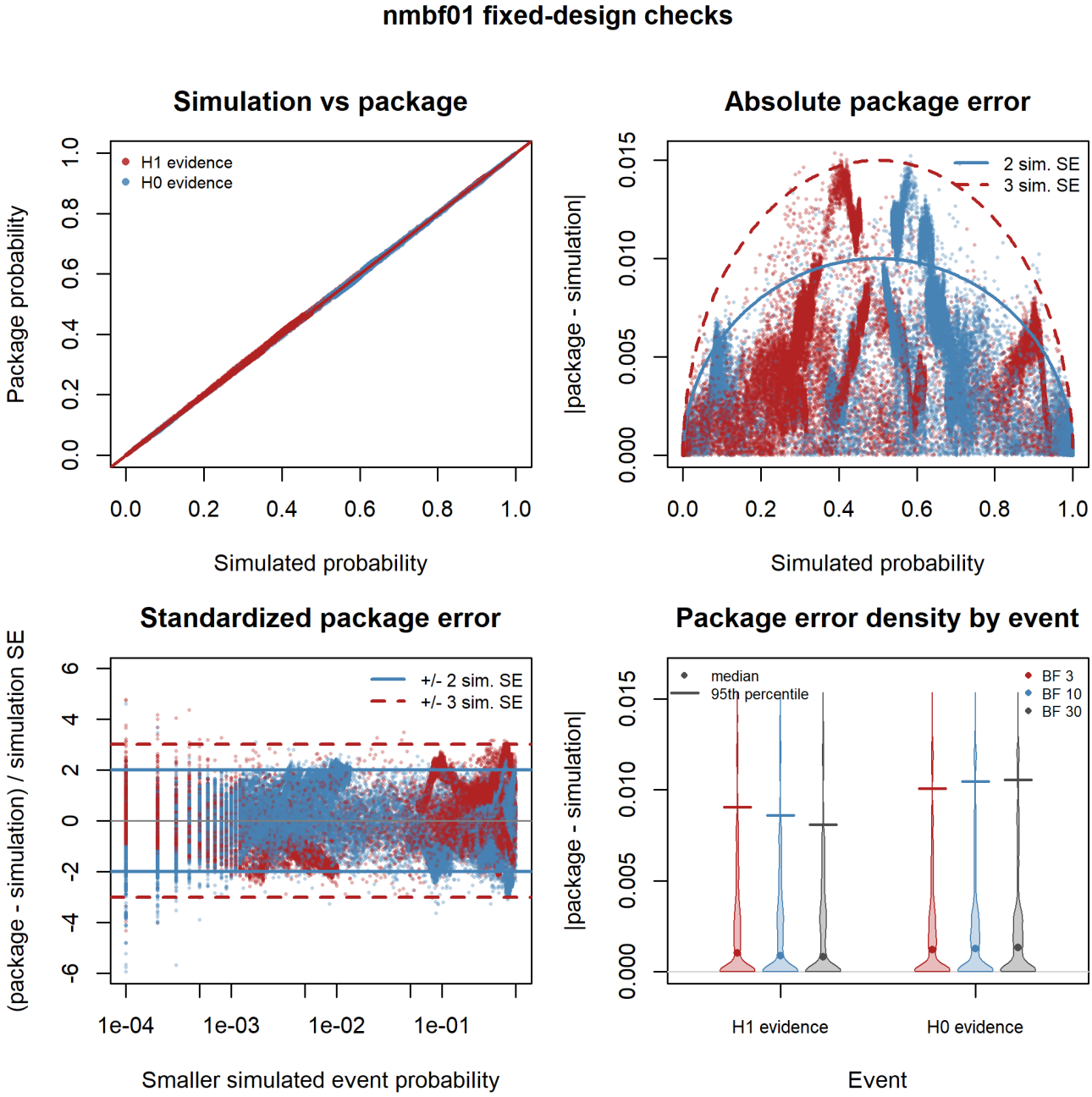
Subscripts + and − denote truncation above and below zero; a nonzero cutoff is shown explicitly. Normal design priors remain untruncated.

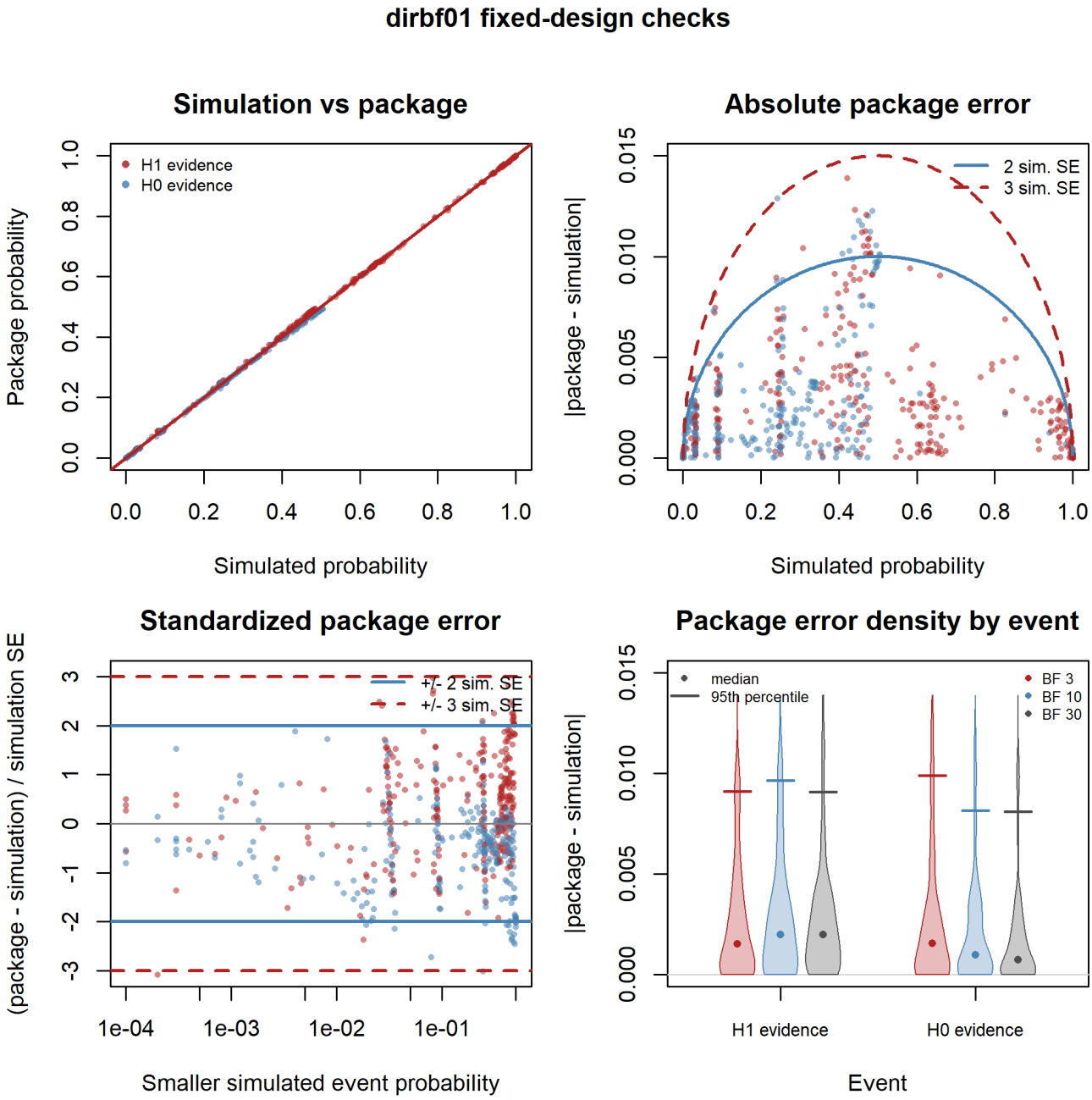
1.1 Fixed-Design

Table 3: Fixed-design verification settings for z-Test analyses. Designs refers to the design-prior table.

Function	Prior/design pairs	Designs	Evidence thresholds	Sample sizes
bf01	51	Z1-Z11	BF 3, 10, 30	30, 130, 260, 380, 500
bf01	22	Z12-Z15	BF 3, 10, 30	510, 2510, 5000, 7500, 10000
nmbf01	33	Z12-Z15	BF 3, 10, 30	10 to 10000 by 10
nmbf01	72	Z1-Z11	BF 3, 10, 30	10 to 500 by 10
dirbf01	16	Z1-Z11	BF 3, 10, 30	30, 130, 260, 380, 500
dirbf01	7	Z12-Z15	BF 3, 10, 30	510, 2510, 5000, 7500, 10000
bf01 (Normal$_+$, Normal$_-$)	70	Z1-Z5, Z7-Z8	BF 3, 10, 30	10 to 500 by 10


bf01 fixed-design checks
Simulation vs package
Package probability
Simulated probability
H1 evidence
H0 evidence
Inconclusive
Absolute package error
|package - simulation|
Simulated probability
2 sim. SE
3 sim. SE
Standardized package error
(package - simulation) / simulation SE
Smaller simulated event probability
+/- 2 sim. SE
+/- 3 sim. SE
Package error density by event
|package - simulation|
Event
median
95th percentile
BF 3
BF 10
BF 30
H1 evidence
H0 evidence
Inconclusive

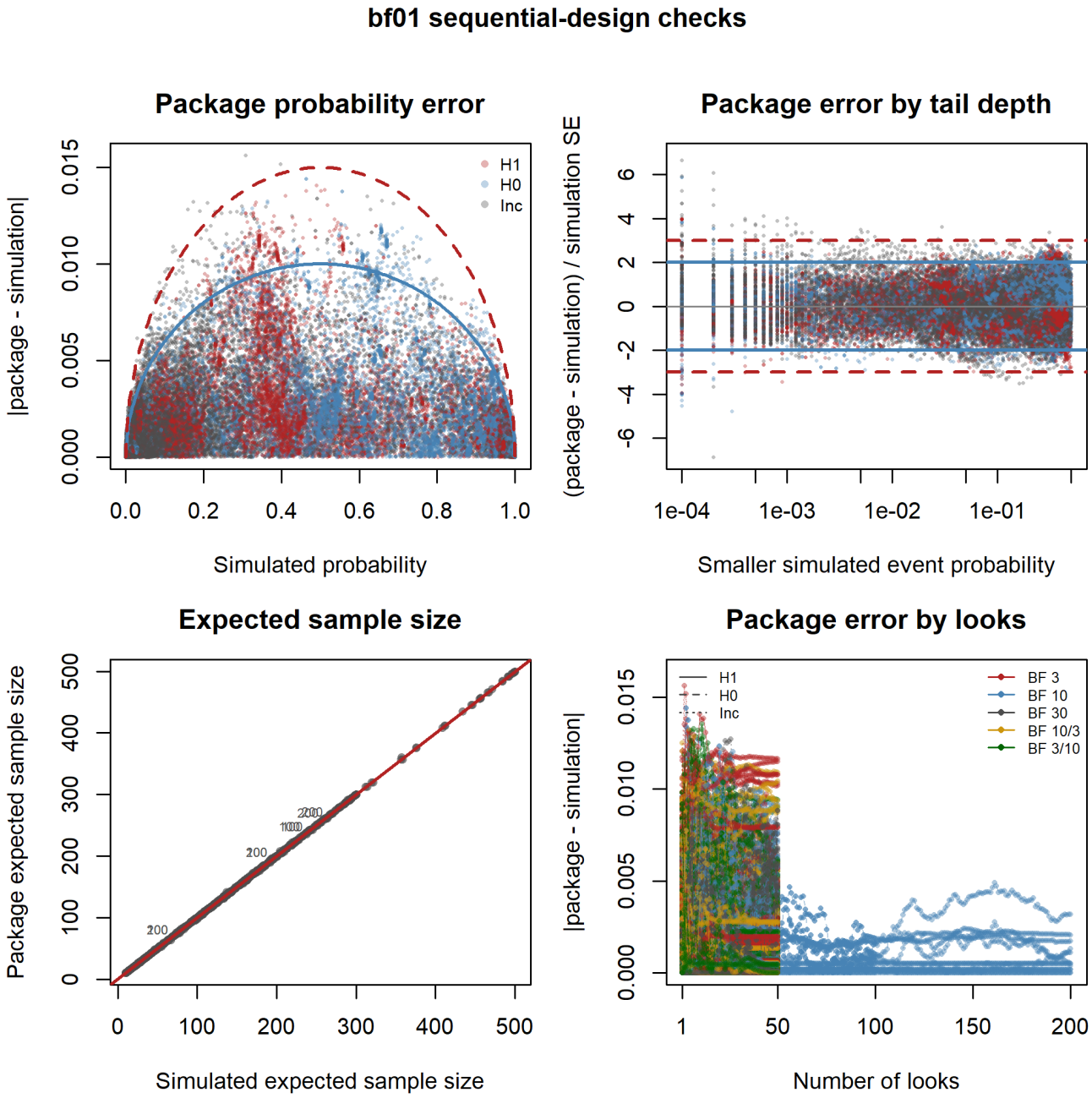

nmbf01 fixed-design checks
Simulation vs package
Package probability
Simulated probability
H1 evidence
H0 evidence
Absolute package error
|package - simulation|
Simulated probability
2 sim. SE
3 sim. SE
Standardized package error
(package - simulation) / simulation SE
Smaller simulated event probability
+/- 2 sim. SE
+/- 3 sim. SE
Package error density by event
|package - simulation|
Event
median
95th percentile
BF 3
BF 10
BF 30
H1 evidence
H0 evidence

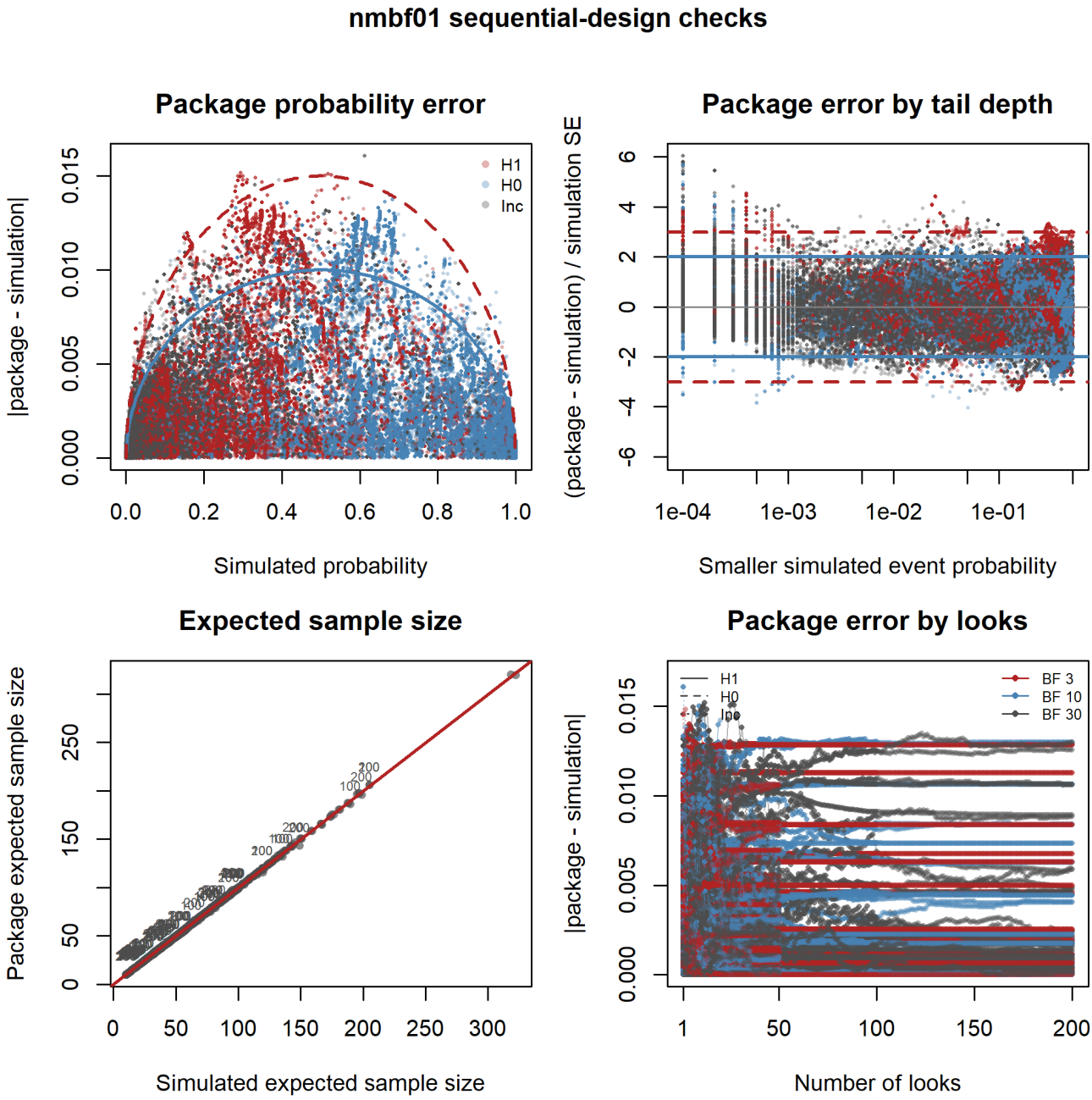


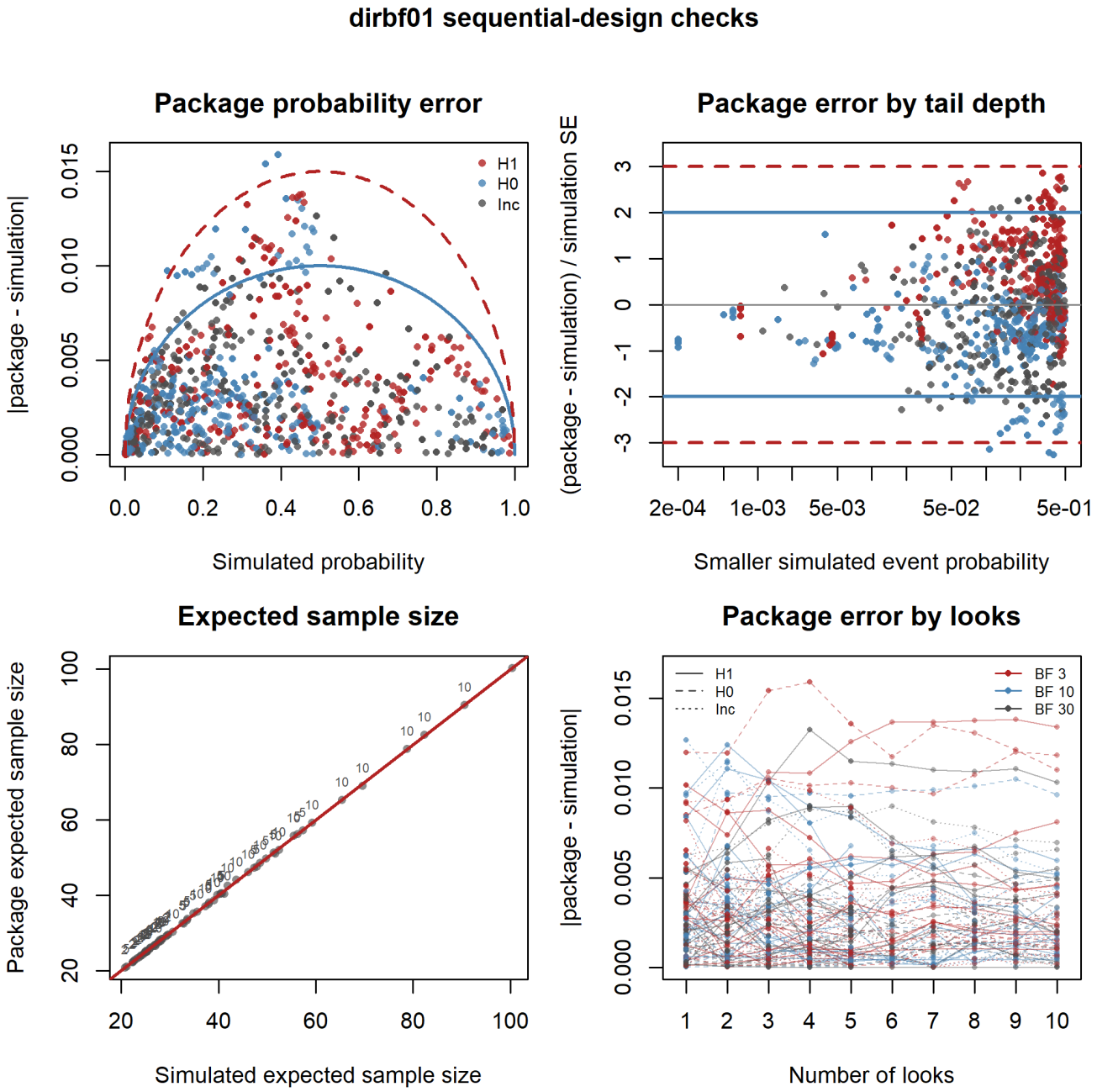
1.2 Sequential-Design

Table 4: Sequential-design verification settings for z-Test analyses. Designs refers to the design-prior table.

Function	Prior/design pairs	Designs	Evidence thresholds	Look sequence
bf01	16	Z4-Z5, Z7, Z11-Z12, Z14	BF 3, 10, 30	start 20, by 10 (2, 5, 10 looks)
nmbf01	86	Z1-Z15	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)
dirbf01	11	Z4-Z5, Z7-Z9, Z11-Z12, Z14	BF 3, 10, 30	start 20, by 10 (2, 5, 10 looks)
bf01 (Normal$_+$, Normal$_-$)	74	Z1-Z5, Z7-Z8, Z12-Z13	BF 3, 10, 30; 10/3, 3/10	1, 3, 50 looks; selected 100, 200 looks


bf01 sequential-design checks
Package probability error
|package - simulation|
0.000
0.005
0.010
0.015
H1
H0
Inc
0.0
0.2
0.4
0.6
0.8
1.0
Simulated probability
Package error by tail depth
(package - simulation) / simulation SE
-6
-4
-2
0
2
4
6
1e-04
1e-03
1e-02
1e-01
Smaller simulated event probability
Expected sample size
Package expected sample size
0
100
200
300
400
500
200
200
1000
2000
0
100
200
300
400
500
Simulated expected sample size
Package error by looks
|package - simulation|
0.000
0.005
0.010
0.015
H1
H0
Inc
BF 3
BF 10
BF 30
BF 10/3
BF 3/10
1
50
100
150
200
Number of looks


nmbf01 sequential-design checks
Package probability error
|package - simulation|
0.000
0.005
0.010
0.015
H1
H0
Inc
0.0
0.2
0.4
0.6
0.8
1.0
Simulated probability
Package error by tail depth
(package - simulation) / simulation SE
-6
-4
-2
0
2
4
6
1e-04
1e-03
1e-02
1e-01
Smaller simulated event probability
Expected sample size
Package expected sample size
0
50
150
250
0
50
100
150
200
250
300
Simulated expected sample size
Package error by looks
|package - simulation|
0.000
0.005
0.010
0.015
H1
H0
Inc
BF 3
BF 10
BF 30
1
50
100
150
200
Number of looks


dirbf01 sequential-design checks
Package probability error
|package - simulation|
0.000
0.005
0.010
0.015
H1
H0
Inc
0.0
0.2
0.4
0.6
0.8
1.0
Simulated probability
Package error by tail depth
(package - simulation) / simulation SE
-3
-2
-1
0
1
2
3
2e-04
1e-03
5e-03
5e-02
5e-01
Smaller simulated event probability
Expected sample size
Package expected sample size
20
40
60
80
100
Simulated expected sample size
Package error by looks
|package - simulation|
H1
H0
Inc
BF 3
BF 10
BF 30
1
2
3
4
5
6
7
8
9
10
Number of looks

2 *t*-Test

Simulated Design Priors

Table 5: Simulated design priors for the t-Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions	Sampling	n2/n1
T1	Point(0)	10 to 500 by 10	10000	two-sample	1
T2	Point(0.5)	10 to 500 by 10	10000	two-sample	1
T3	Normal(0.5, 0.1)	10 to 500 by 10	10000	two-sample	1
T4	Normal(0.5, 0.5)	10 to 500 by 10	10000	two-sample	1
T5	Point(0)	10 to 500 by 10	10000	one-sample	
T6	Point(0.5)	10 to 500 by 10	10000	one-sample	
T7	Point(0)	10 to 500 by 10	10000	paired	
T8	Point(0.5)	10 to 500 by 10	10000	paired	
T9	Point(-0.4)	10 to 500 by 10	10000	two-sample	1
T10	Point(0)	10 to 500 by 10	10000	two-sample	1.1
T11	Point(0.35)	10 to 500 by 10	10000	two-sample	1.1
T12	Normal(0, 0.5)	10 to 500 by 10	10000	two-sample	1
T13	Point(0)	10 to 10000 by 10	10000	two-sample	1
T14	Point(0.2)	10 to 10000 by 10	10000	two-sample	1
T15	Normal(0.2, 0.1)	10 to 10000 by 10	10000	two-sample	1
T16	Normal(0.2, 0.5)	10 to 10000 by 10	10000	two-sample	1

Analysis Priors

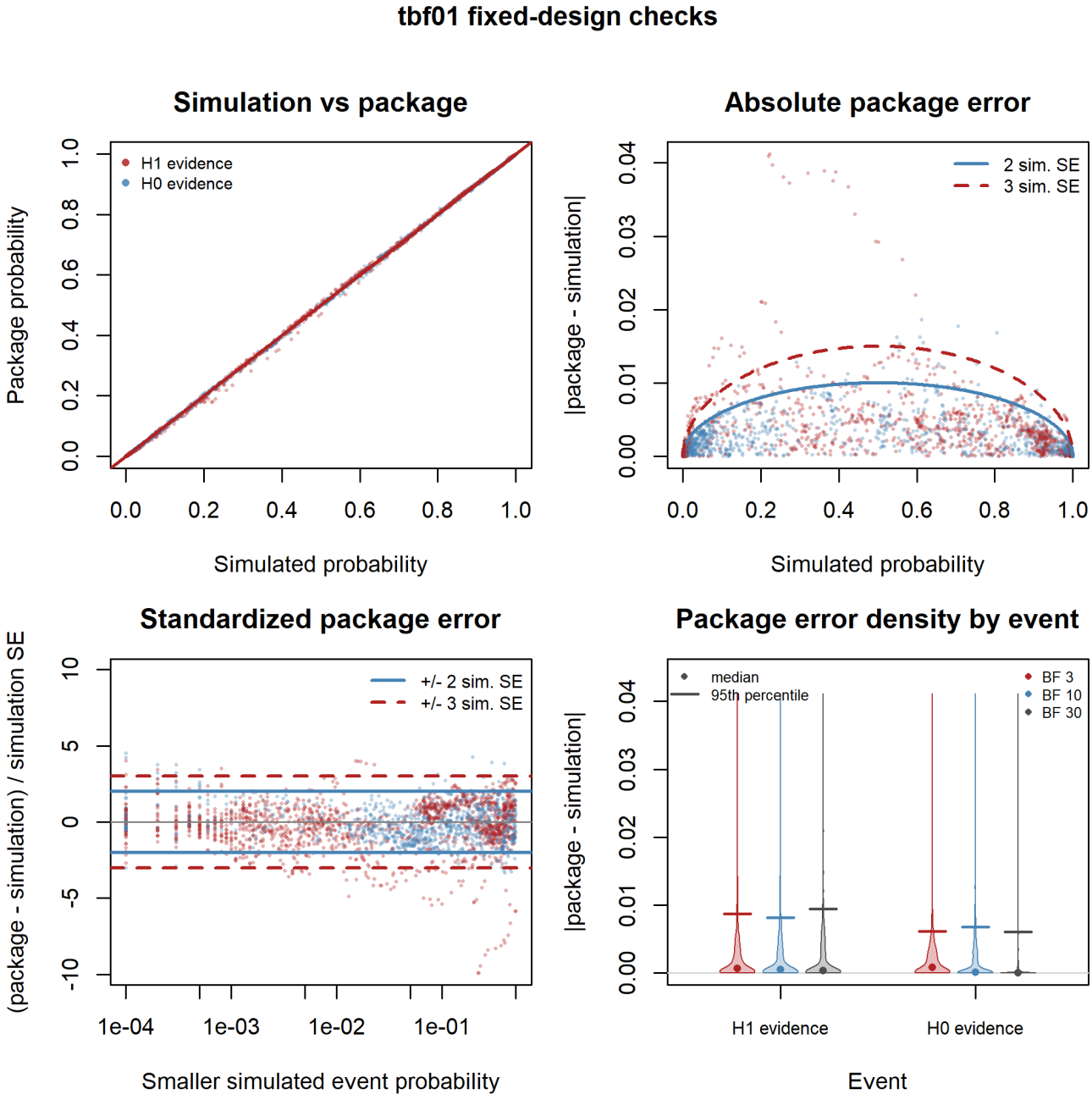
Table 6: Analysis priors used for the t-Test verification run. Designs lists the matching short IDs from the design-prior table.

Function	H1	H0	Designs
tbf01	Cauchy$_+$(0, 0.707)	Point(0)	T1-T16
tbf01	Cauchy$_-$(0, 0.707)	Point(0)	T1-T16
tbf01	Cauchy(0, 0.35)	Point(0)	T1-T16
tbf01	Cauchy(0, 0.707)	Point(0)	T1-T16
tbf01	Cauchy(0, 1)	Point(0)	T1-T16
tbf01	StudentT$_+$(3, 0.5, 0.1)	Point(0)	T2-T4, T6, T8, T11, T14-T16
tbf01	StudentT$_+$(3, 0.5, 0.35; $>$ 0.2)	Point(0.2)	T11, T14-T16
tbf01	StudentT$_+$(30, 0.5, 0.35; $>$ 0.2)	Point(0.2)	T1, T5, T7, T10, T13
tbf01	StudentT(3, 0, 0.707)	Point(0)	T1-T16
tbf01	StudentT(30, 0, 0.707)	Point(0)	T1-T16

2.1 Fixed-Design

Table 7: Fixed-design verification settings for t-Test analyses. Designs refers to the design-prior table.

Function	Prior/design pairs	Designs	Evidence thresholds	Sample sizes
tbf01	95	T1-T12	BF 3, 10, 30	30, 130, 260, 380, 500
tbf01	35	T13-T16	BF 3, 10, 30	510, 2510, 5000, 7500, 10000

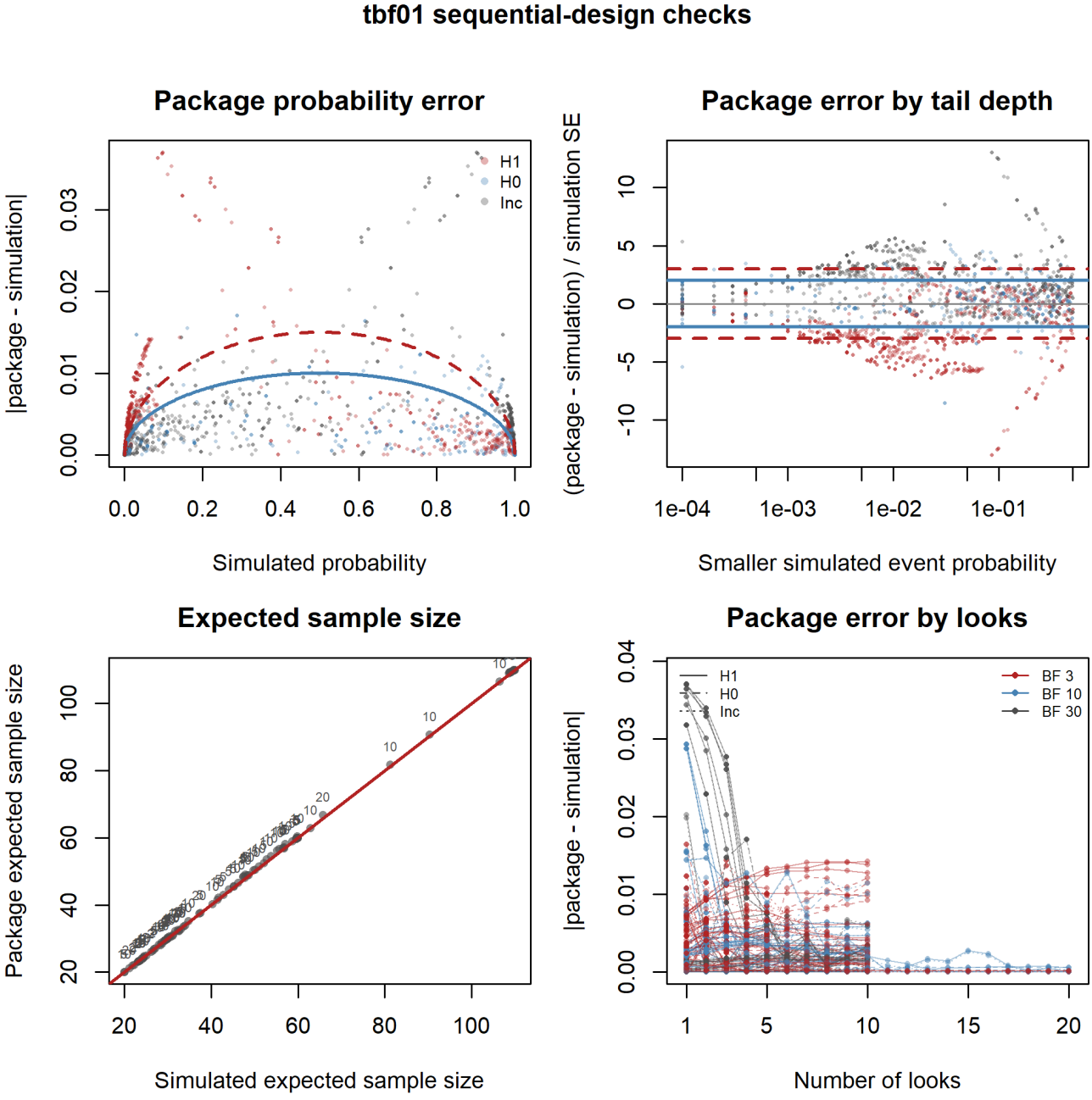


2.2 Sequential-Design

Table 8: Sequential-design verification settings for t-Test analyses. Designs refers to the design-prior table.

Function	Prior/design pairs	Designs	Evidence thresholds	Look sequence
tbf01	27	T2, T5-T6, T9, T13	BF 3, 10, 30	start 20, by 10 (2, 5, 10, 20 looks)

The sequential t-test probabilities use a normal approximation to the sampling distribution of the test statistics. Comparisons with exact-t simulations therefore diagnose this approximation; they do not certify exact-t accuracy. Numerical validity and fixture integrity remain required checks.



3 Binomial Test

Simulated Design Priors

Table 9: Simulated design priors for the Binomial Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions
B1	Point(0.5)	10 to 500 by 10	10000
B2	Point(0.6)	10 to 500 by 10	10000
B3	Point(0.2)	10 to 500 by 10	10000
B4	Point(0.75)	10 to 500 by 10	10000
B5	Beta(60, 40)	10 to 500 by 10	10000
B6	$\text{Beta}_+(1, 1; > 0.5)$	10 to 500 by 10	10000
B7	$\text{Beta}_-(1, 1; \leq 0.5)$	10 to 500 by 10	10000
B8	Point(0.5)	10 to 10000 by 10	10000
B9	Point(0.55)	10 to 10000 by 10	10000

Analysis Priors

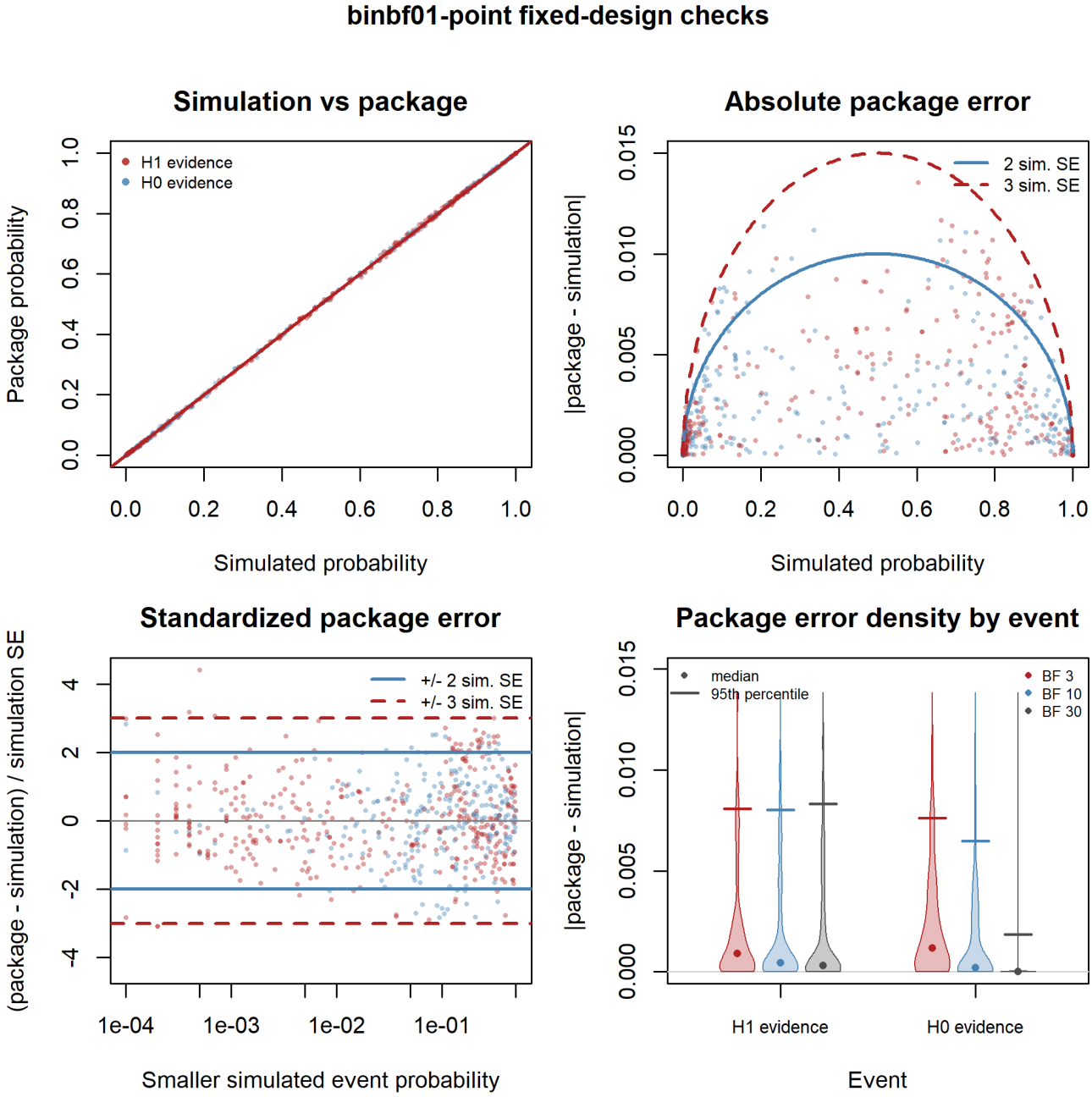
Table 10: Analysis priors used for the Binomial Test verification run. Designs lists the matching short IDs from the design-prior table.

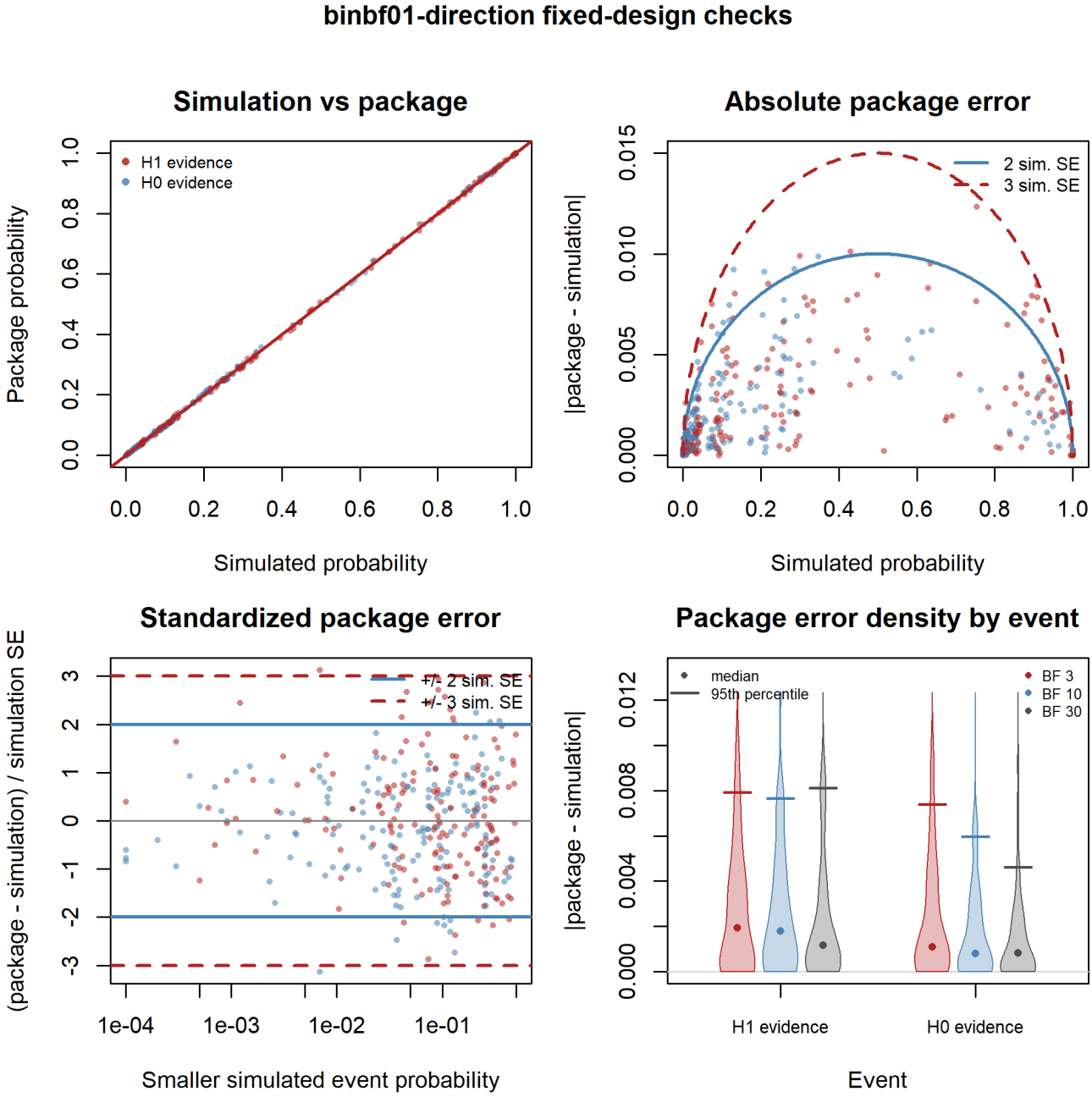
Function	H1	H0	Designs
binbf01 (direction)	$\text{Beta}_+(1, 1; > 0.2)$	$\text{Beta}_-(1, 1; \leq 0.2)$	B3
binbf01 (direction)	$\text{Beta}_+(1, 1; > 0.5)$	$\text{Beta}_-(1, 1; \leq 0.5)$	B1-B9
binbf01 (direction)	$\text{Beta}_+(1, 1; > 0.55)$	$\text{Beta}_-(1, 1; \leq 0.55)$	B9
binbf01 (direction)	$\text{Beta}_+(1, 1; > 0.6)$	$\text{Beta}_-(1, 1; \leq 0.6)$	B2, B5
binbf01 (direction)	$\text{Beta}_+(1, 1; > 0.75)$	$\text{Beta}_-(1, 1; \leq 0.75)$	B4
binbf01 (point)	Beta(1, 1)	Point(0.2)	B3
binbf01 (point)	Beta(1, 1)	Point(0.5)	B1-B9
binbf01 (point)	Beta(1, 1)	Point(0.55)	B9
binbf01 (point)	Beta(1, 1)	Point(0.6)	B2, B5
binbf01 (point)	Beta(1, 1)	Point(0.75)	B4
binbf01 (point)	Beta(40, 60)	Point(0.5)	B1-B9
binbf01 (point)	Beta(5100, 4900)	Point(0.5)	B8-B9
binbf01 (point)	Beta(60, 40)	Point(0.5)	B1-B9

3.1 Fixed-Design

Table 11: Fixed-design verification settings for Binomial Test analyses. Designs refers to the design-prior table.

Function	Prior/design pairs	Designs	Evidence thresholds	Sample sizes
binbf01 (point)	25	B1-B7	BF 3, 10, 30	30, 130, 260, 380, 500
binbf01 (point)	9	B8-B9	BF 3, 10, 30	510, 2510, 5000, 7500, 10000
binbf01 (direction)	11	B1-B7	BF 3, 10, 30	30, 130, 260, 380, 500
binbf01 (direction)	3	B8-B9	BF 3, 10, 30	510, 2510, 5000, 7500, 10000

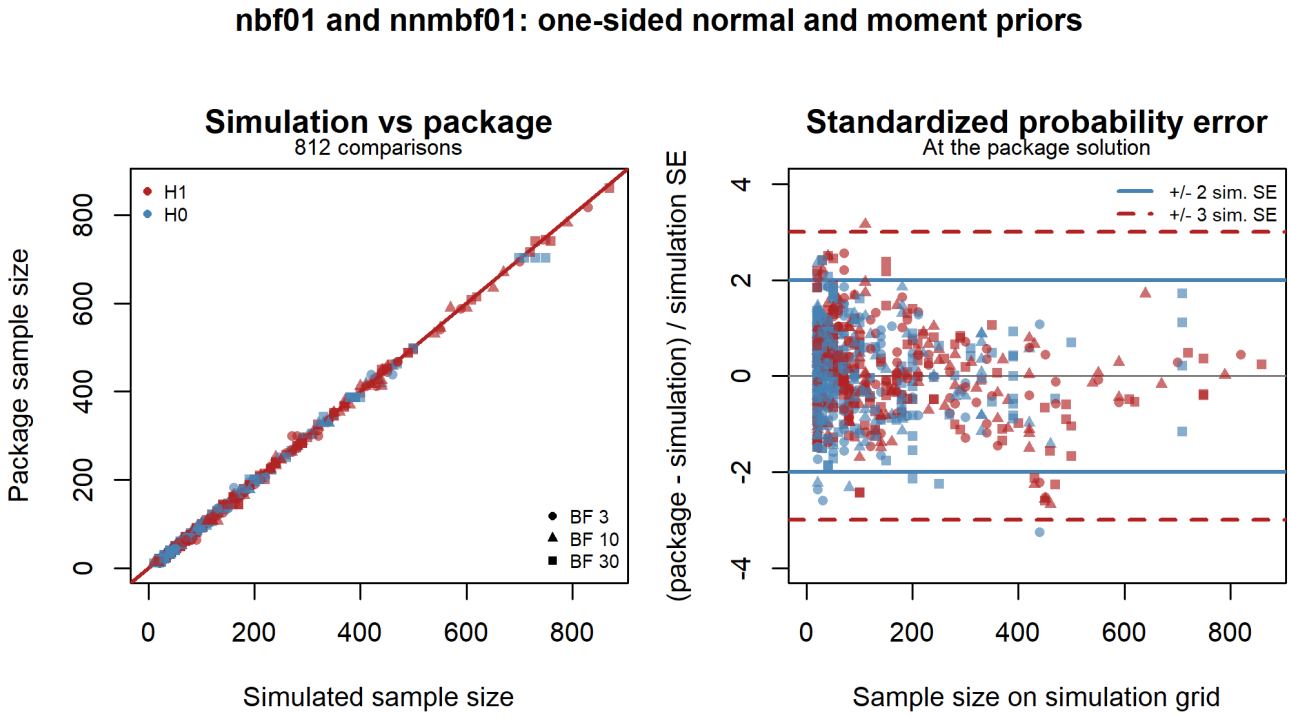

binbf01-point fixed-design checks
Simulation vs package
Package probability
Simulated probability
H1 evidence
H0 evidence
Absolute package error
|package - simulation|
Simulated probability
2 sim. SE
3 sim. SE
Standardized package error
(package - simulation) / simulation SE
Smaller simulated event probability
+/- 2 sim. SE
+/- 3 sim. SE
Package error density by event
|package - simulation|
Event
median
95th percentile
BF 3
BF 10
BF 30
H1 evidence
H0 evidence


binbf01-direction fixed-design checks
Simulation vs package
Package probability
Simulated probability
H1 evidence
H0 evidence
Absolute package error
|package - simulation|
Simulated probability
2 sim. SE
3 sim. SE
Standardized package error
(package - simulation) / simulation SE
Smaller simulated event probability
+/- 2 sim. SE
+/- 3 sim. SE
Package error density by event
|package - simulation|
Event
median
95th percentile
BF 3
BF 10
BF 30
H1 evidence
H0 evidence

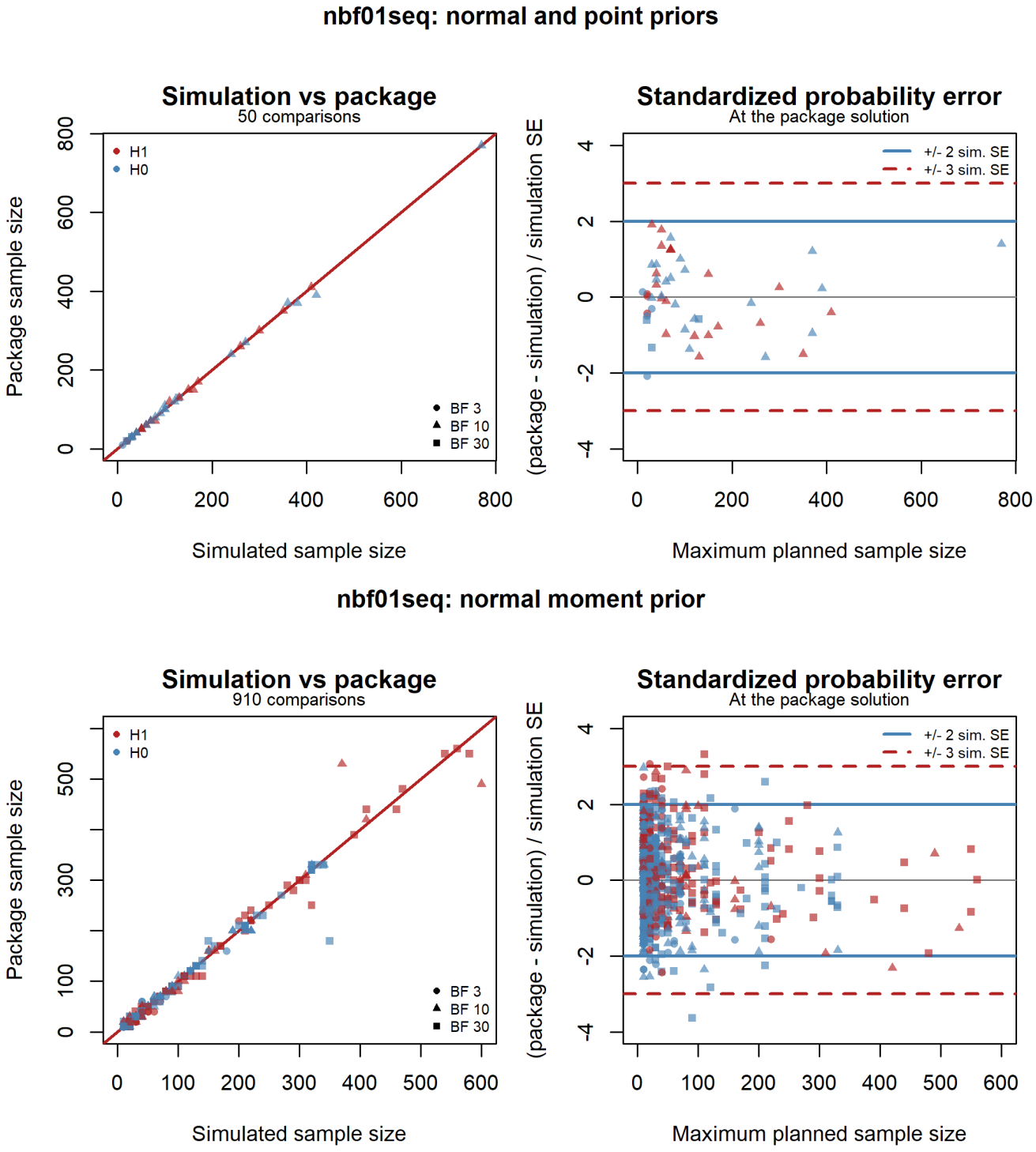
4 Sample-Size Search Checks

The searches cover `nbf01` with one-sided Normal priors, `nnmbf01`, `nbf01seq`, and `ntbf01seq`. They compare the first sample sizes at which simulated and computed probabilities of evidence for H1 or H0 reach a target. Sequential sample sizes refer to the maximum planned sample size. Only finite results from both methods enter the figures.

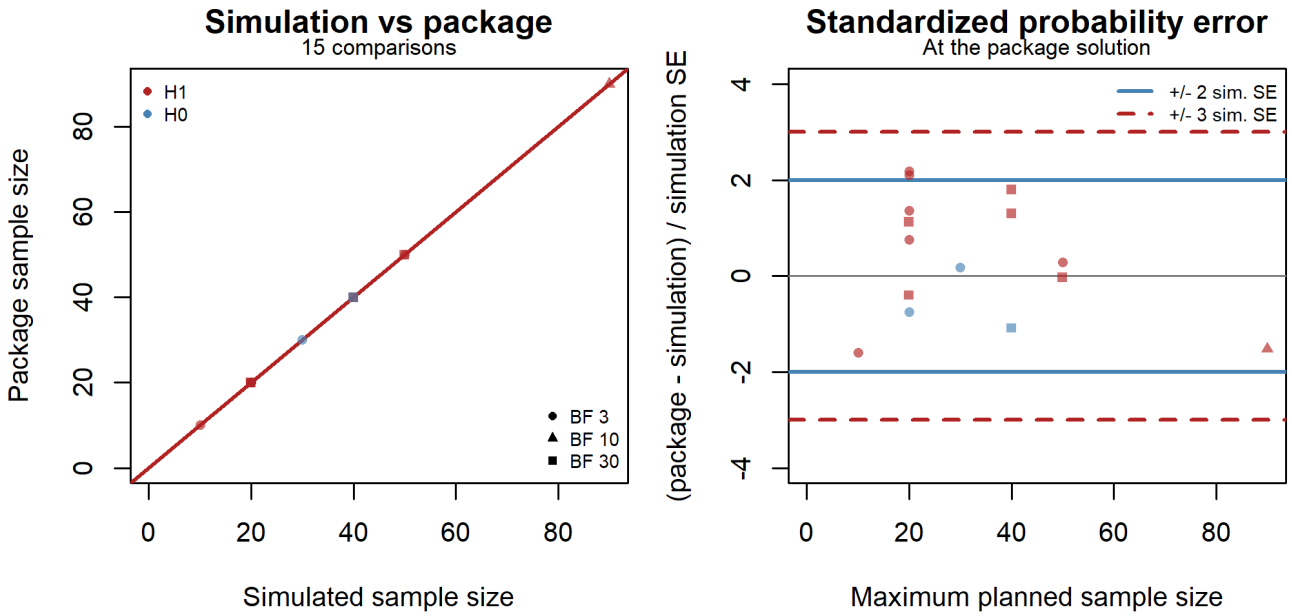
4.1 Fixed-Design z-Test: `nbf01`, `nnmbf01`



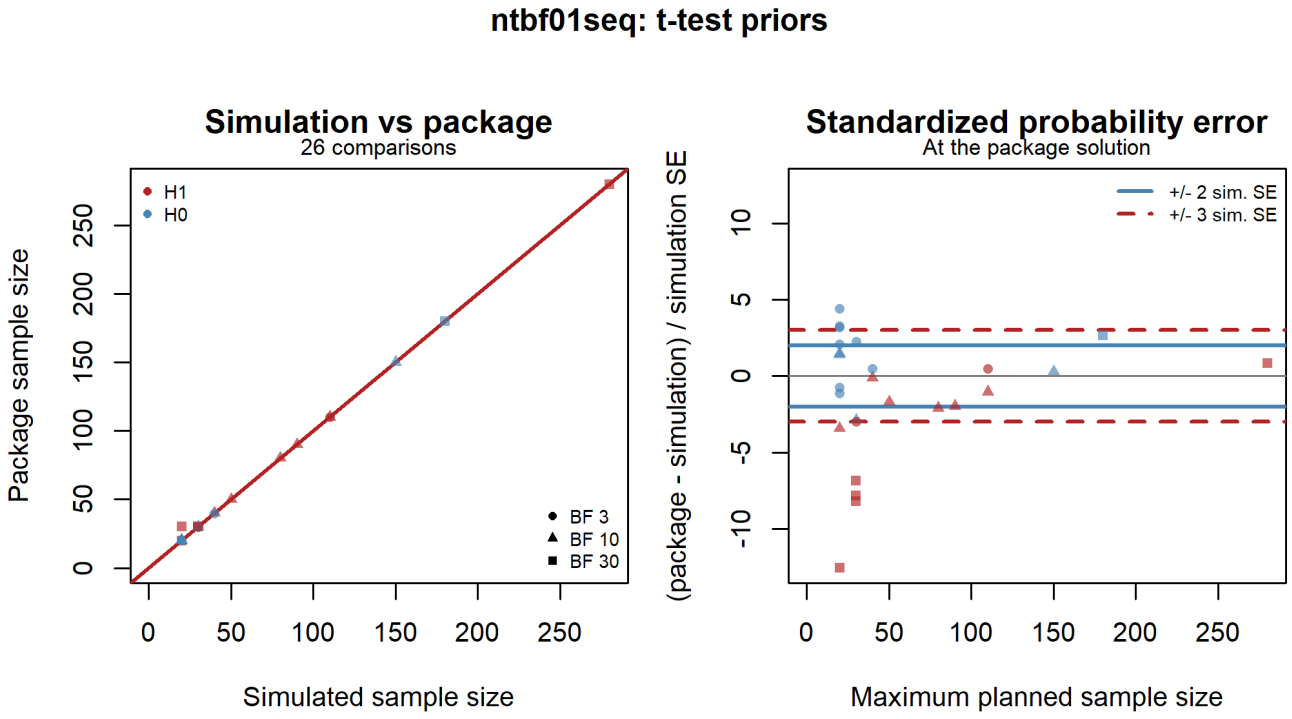
4.2 Sequential z-Test: `nbf01seq`



nbf01seq: directional hypotheses



4.3 Sequential t-Test: `ntbf01seq`



5 Run Time Diagnostics

Elapsed times are recorded for the package calculations shown above. Fixed-design probability calls can cover several rows at once because those functions are vectorized; most sequential and sample-size search calls are one setting at a time.

Table 12: Probability and sample-size search run times. Fixed probability rows can be vectorized, so calls and rows covered need not match.

Function	Calls	Rows covered	Median	P90	Max	Total
pbf01	73	2,190	0.002 s	0.004 s	0.073 s	0.316 s
pbf01seq	239	906	0.102 s	1.189 s	32.938 s	9.009 min
pbinbf01	48	1,440	0.036 s	0.064 s	0.271 s	2.383 s
pnmbf01	35	1,050	0.002 s	0.002 s	1.544 s	1.61 s
ptbf01	130	3,900	0.883 s	1.773 s	1.902 s	1.886 min
ptbf01seq	148	148	0.567 s	1.582 s	32.745 s	5.143 min
nbf01seq	50	50	0.006 s	0.164 s	35.794 s	42.282 s
ntbf01seq	26	26	0.117 s	1.562 s	2.9 s	12.44 s

Table 13: One-sided z-test run times. Each fixed pbf01 run times the H1 and H0 vector calls together; each other run times one package call.

Function	Timed runs	Median	P90	Max	Total
nbf01	840	0.007 s	0.012 s	0.085 s	7.017 s
nbf01seq	2212	5.194 s	22.052 s	29.292 s	229.005 min
nnmbf01	2100	0.005 s	0.013 s	0.073 s	14.787 s
pbf01	210	0.018 s	0.043 s	0.151 s	5.173 s
pbf01seq	1828	0.013 s	3.543 s	1.593 min	107.537 min
pnmbf01	1072	0.001 s	0.208 s	2.198 s	1.729 min

Histograms show the calls in the probability and sample-size search timing table.

